

The whole thesis as one story

What happens in a shipyard, what the research asks, what has been done, what is under way and what comes next.

Road-occupying transport in shipyards: network-bound capacity, deadlock-free orchestration and fleet-network co-design

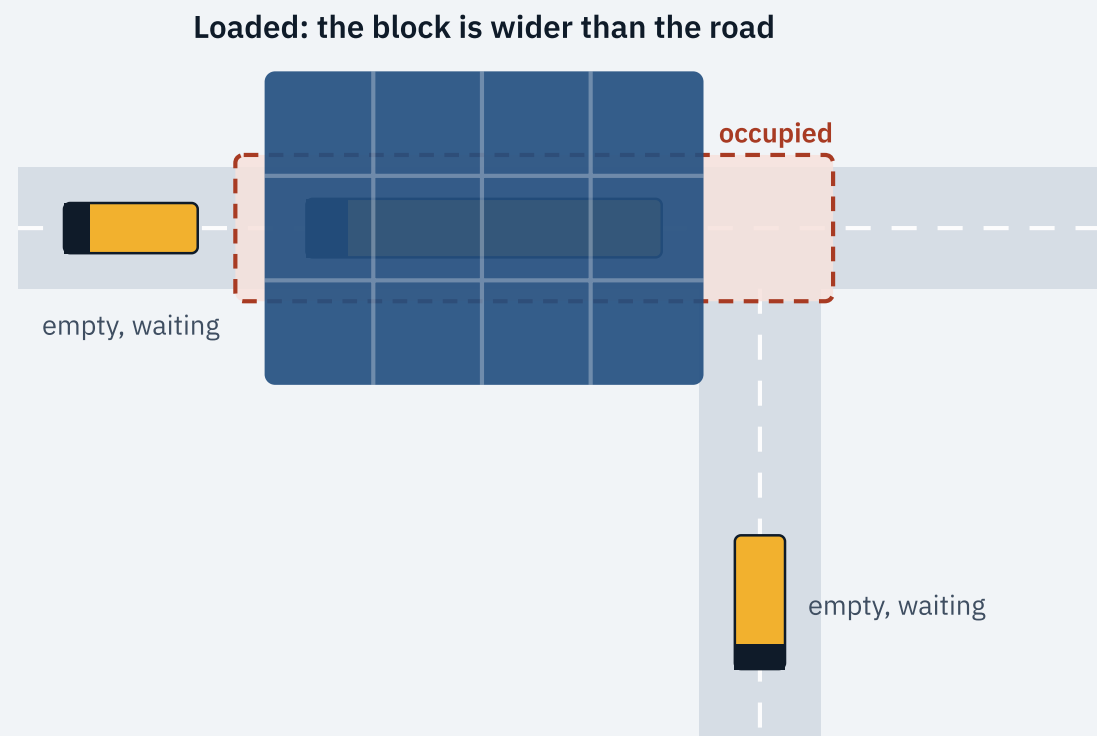
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Full results in the thesis overview

The usual model: vehicles never meet

Reality: a loaded vehicle owns the road



The order of this briefing (click to jump)

1

What happens in a shipyard

Blocks are wider than roads: a loaded vehicle takes the road

p. 3

2

What the research asks

Where the usual model goes wrong; the question, five claims, the capacity interval

p. 5

3

What has been done

Flag experiment, two yards, three corrections, the dock mouth

p. 8

4

What is under way

Where deadlock comes from; a smarter rule that never jams

p. 13

5

What comes next

Vehicles or roads; more yards; theorems and the first paper

p. 15

6

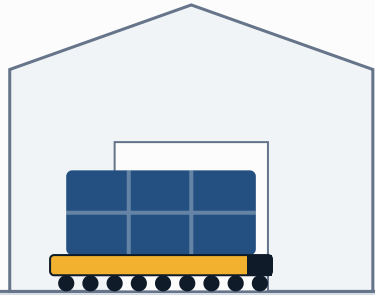
The whole story in one figure

What each chapter does on that figure

p. 16

A ship is built like a stack of blocks: blocks, transporters, docks and cranes

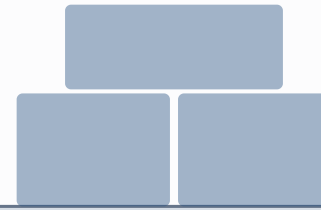
① Shop: build blocks



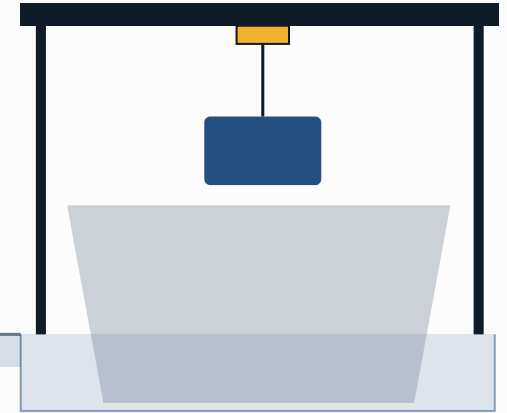
② Stockyard: store



③ Assembly: join into units



④ Dock: lift onto hull



1

Blocks

The hull is built in pieces called blocks: hundreds of tonnes each, often 15–40 m long and wide.

2

Transporters

Hydraulic platform transporters carry each block between shops, stockyards and assembly areas several times. An empty one is about 7 m wide.

3

Docks and cranes

Finally a block goes to the dock, where a gantry crane lifts it onto the hull. When the cranes are busy, vehicles queue at the dock.

4

The key fact

Once loaded, vehicle and block are wider than the road: the whole segment and junction are taken, and oncoming vehicles must wait.

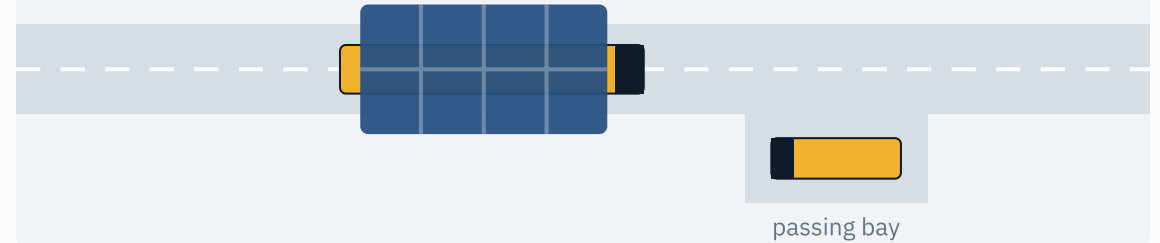
The usual model treats vehicles as points that never meet

Free-flow model: the two pass through each other



Travel times from a distance table, no constraint between vehicles

Reality: the empty one waits in a bay until the road is free



Earlier studies: free flow

Vehicles never affect each other; more vehicles always move more, at most with shrinking gains. Like cars on an empty highway.

Reality: occupancy

A loaded vehicle holds the segment and the junction alone; oncoming vehicles wait, with no passing and no overtaking. More like a train on a single track.

Researchers in the field have named this gap themselves: Sun et al. (2024) write in their conclusion that some roads take only one transporter at a time, and leave it as future work. In 288 delay records from a Shanghai yard, transporters occupy several lanes and other vehicles must give way (Wang et al. 2024).

Change the premise and four things change with it

1

The ceiling is the network

There is only so much road; extra vehicles just queue. How much moves each day is set, in the end, by the network.

2

More vehicles can be slower

Extra vehicles waiting on the road block everyone else.

3

Dispatching becomes traffic ordering

Scheduling shifts from handing out jobs to deciding who passes which road first, as in railway dispatching.

4

A road change can beat a vehicle

One passing bay, or splitting a road into sections, may be worth several vehicles.

In a shipyard, a loaded transporter does not drive on the road; it owns the road. Capacity is capped by the network, not the fleet.

The question, and five bold answers

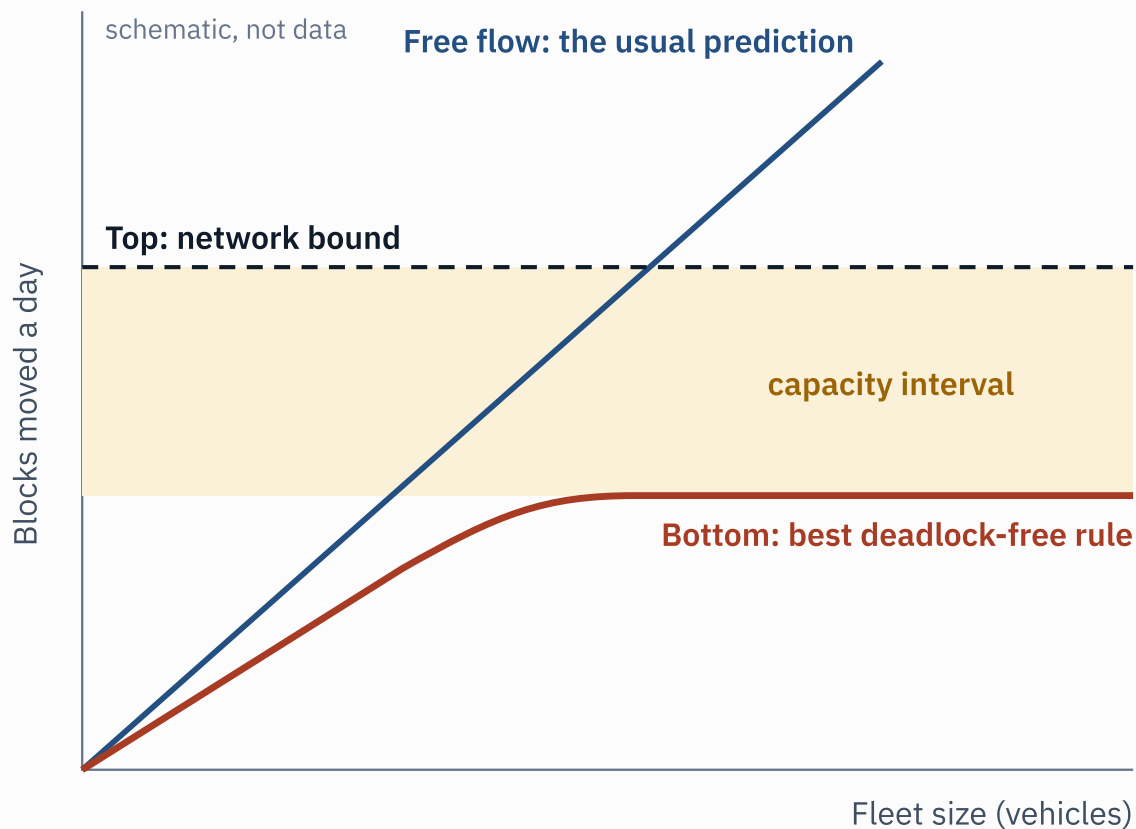
The question: when loaded vehicles own the road, what decides how many blocks a yard can move a day, and how should buying vehicles, scheduling and changing roads be designed together?

CLAIM	WHAT IT SAYS
Two ceilings	Daily output is the smaller of what the fleet can run and what the network can let through; the second can be computed by linear programming.
Trapezoid law	As vehicles are added, output first rises in a straight line, then levels off; if the extra vehicles wait on the road, it can even fall.
From dispatching to traffic ordering	Once output levels off, how much moves depends mostly on release order, where vehicles wait and how road segments are booked.
Road–vehicle exchange rate	When the plateau nears the ceiling, how many vehicles a key road change is worth can be computed.
Transfer across yards	These patterns hold across yards and can be predicted from the map.

Seven hypotheses were written down in advance, each with a result either way: if one holds we gain a finding, if not a useful boundary.

Source: this study's claims and hypotheses (summarised here; the formal wording is fixed in writing).

Capacity is an interval: the network sets the top, scheduling the bottom



Top: the network bound

The most the road network itself can let through, whatever scheduling rule is used.

Bottom: the best rule that never jams

What the best rule found so far achieves while guaranteeing that vehicles never lock each other in.

The gap says what to do

Bottom far below the top: improve scheduling first (chapter 4).

Bottom close to the top: only road changes help (chapter 5).

Step 1: the flag experiment shows the effect is real (Okpo yard)

On the real road network of Hanwha Ocean's Okpo yard (centrelines traced from public imagery), each road segment and junction is a resource for one vehicle at a time, and a whole day of transport is simulated. Three rules for conflicts are compared:

Free flow

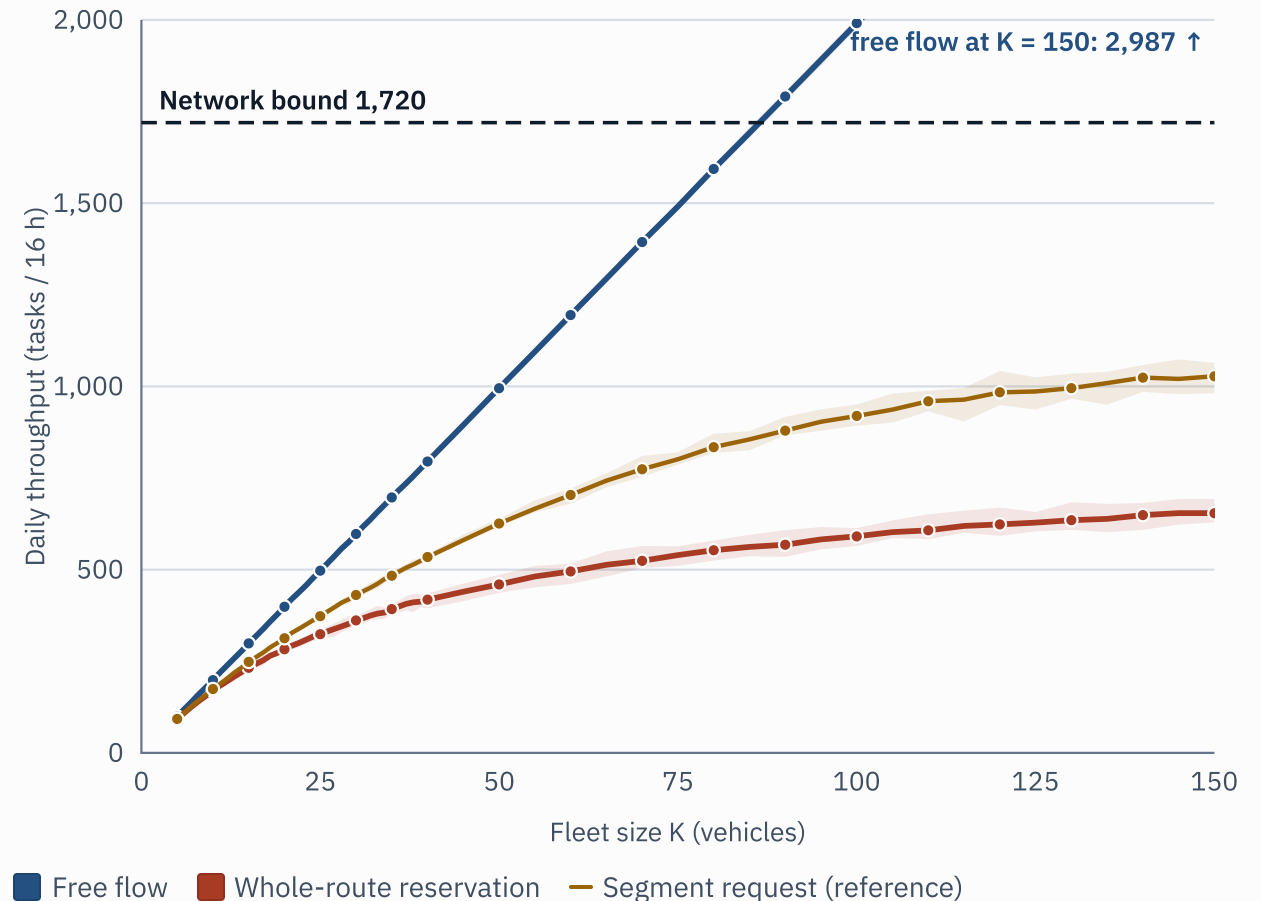
Vehicles never interact: the usual model.

Whole-route reservation

Book the entire route before leaving, as a train is given its path. It never jams, but it is cautious.

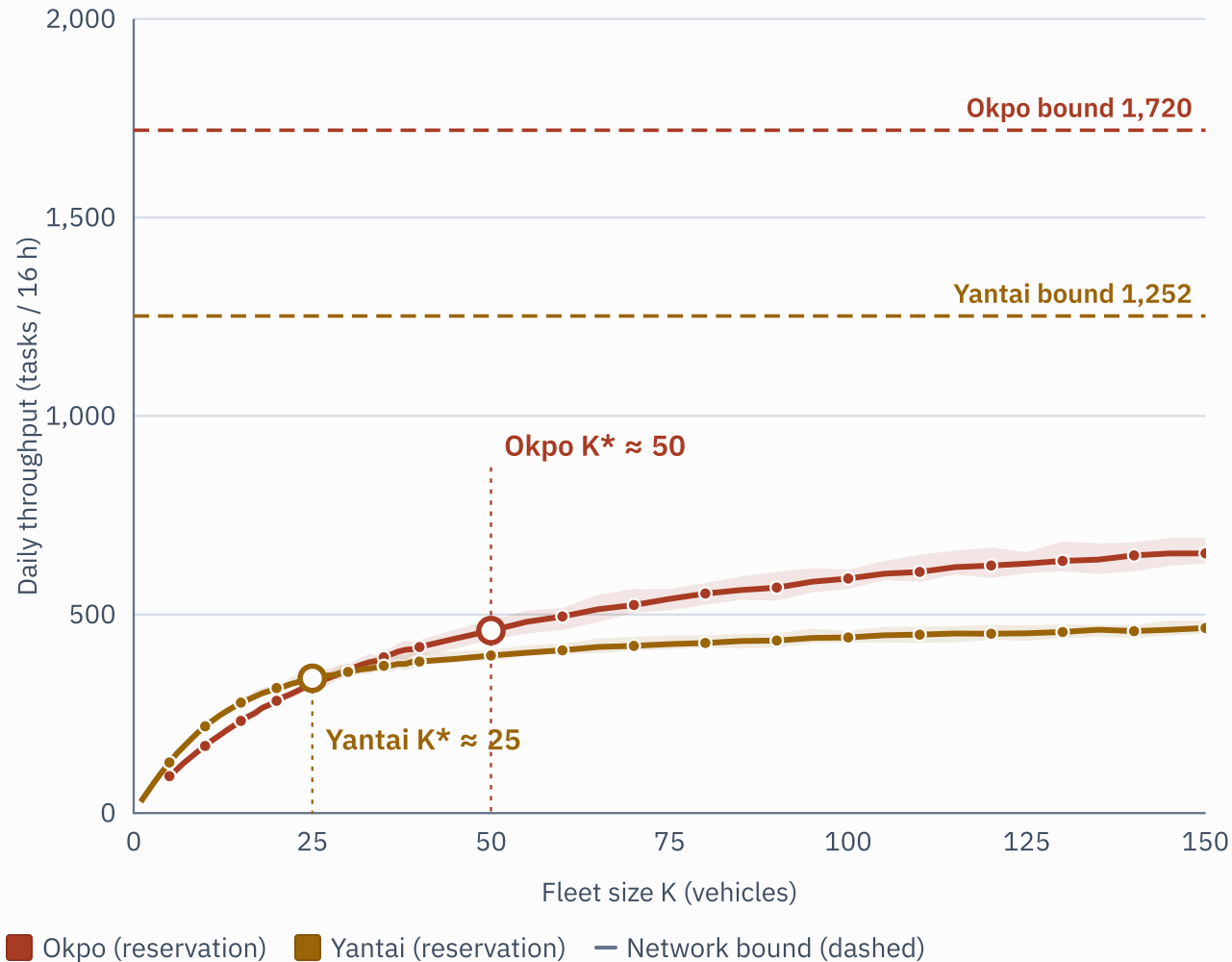
Segment request

Ask for one segment at a time: more efficient, but two vehicles can lock each other in a deadlock. Deadlocks were cleared by teleporting a vehicle out, so this rule is a reference only.



At K = 80: free flow 1,593, reservation only 553, 35% of it

Step 2: the ceiling, a second yard, the dock mouth



Pinning down the ceiling

Tightened network bound (dashed): about 1,720 a day at Okpo, 1,250 at Yantai; none of 50,000+ simulated points exceeds it. Each road also gets a shadow price.

A second yard: Yantai

Both level off, but network limits begin at different fleet sizes (dots): about 50 vehicles at Okpo, 25 at Yantai. Same pattern, layout-specific numbers.

Cranes at the dock mouth

Busier cranes, longer dock queues; past a threshold, yard-wide capacity drops sharply. Step 3 shows why.

Data: simulations and bound computations in this study; saturated demand, mean of 10 seeds; K^* = where one more vehicle first adds under 20% of its free-flow gain. System bound without sampling noise: 1,736 (Okpo), 1,298 (Yantai).

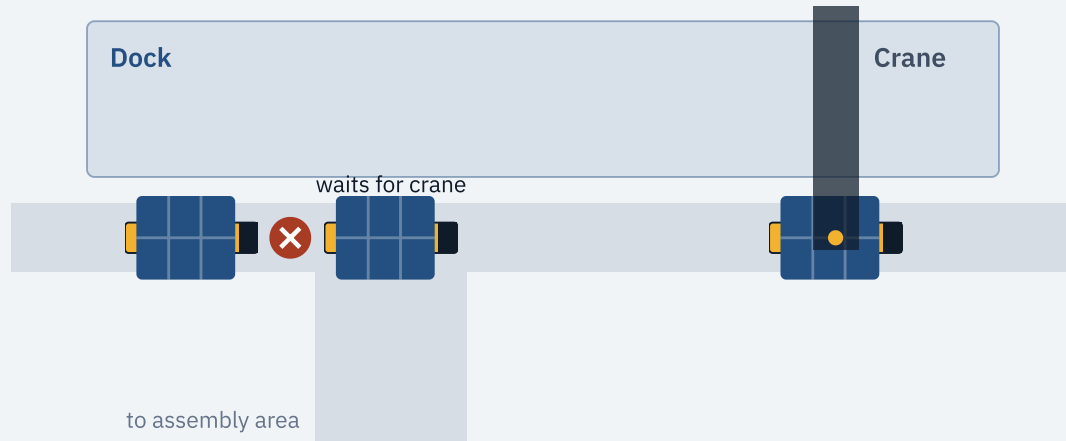
Step 3: a full review of the results, and three corrections

PROBLEM	THE FIX	WHAT CAME OUT
Task volume too small: 97 or 194 a day was a four-vehicle subset	Whole-yard volume converted from public steel use	Okpo moves about 339 blocks a day, within 3% of the 350 Hanwha has published
The sharp drop at the dock relied on teleporting, which is not physical	A safe release rule that never teleports and never deadlocks (like a bank that lends only when every borrower can still finish); the dock road split at its stops	It still drops without teleporting; it does not drop when vehicles wait off the road; splitting the road moves the threshold later. The drop is real, and it is about where vehicles wait
Capacity should be an interval, not one number	[best deadlock-free rule, network bound]	Okpo [627, 1,720]: the bottom is about 36% of the top; scheduling decides the gap, which chapter 4 aims to win back

Sources: corrections after the full review; simulations in this study (physical rules and whole-yard volumes, system bound); external anchors and a literature check; all verified.

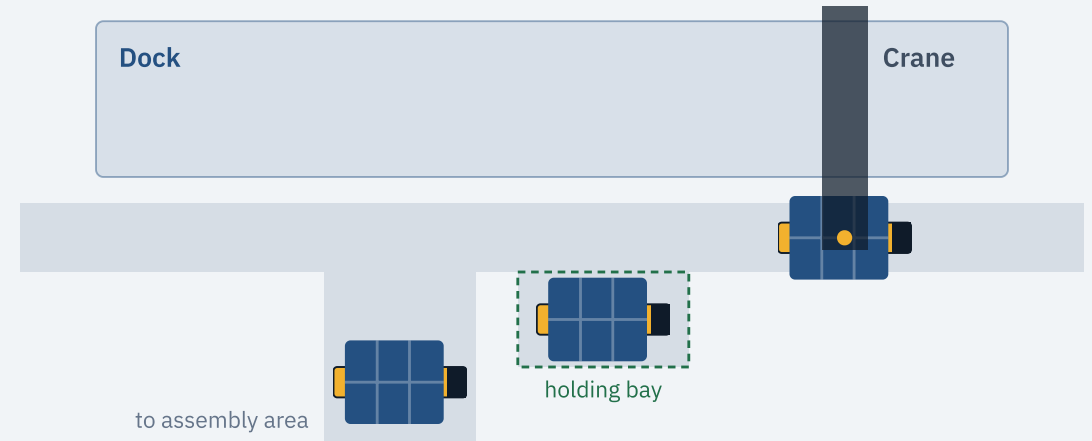
Dock mouth: where vehicles wait decides yard capacity

Waiting for the crane on the dock road



The vehicle waiting for the crane holds the dock road, so traffic to the assembly area is stuck and the queue spreads yard-wide, teleporting or not.

Waiting in an off-road holding bay



The waiting vehicle stands off the road, the way stays open and the queue stays at the dock. Splitting the dock road at its stops does the same.

85% · 60-70%

Crane-use thresholds (Okpo · Yantai): past them, capacity drops sharply.

≈ 88%

Okpo's crane use from public output figures (78–97%), at the threshold.

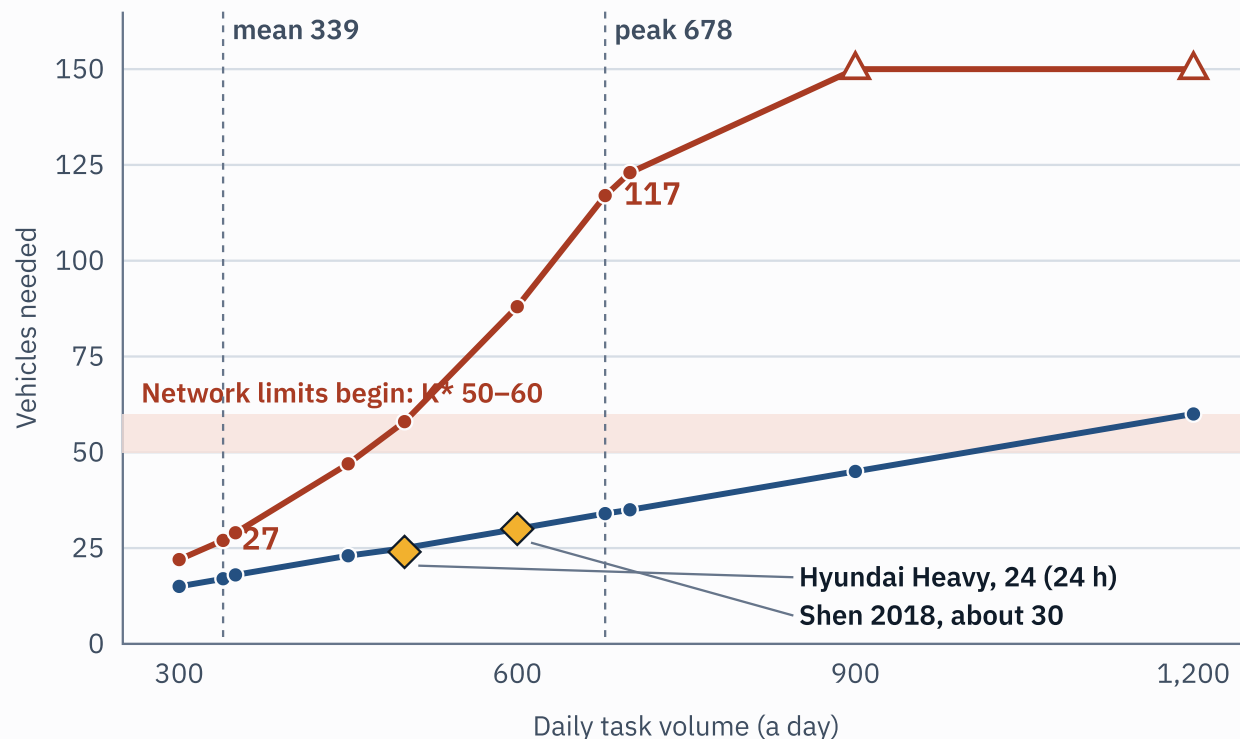
Later threshold

Waiting off the road, or splitting the dock road, lets more vehicles work at once.

What the corrections showed: vehicles needed, and where it binds

16-h day

24-h day



■ Free flow ■ Whole-route reservation ■ Field fleets ■ Network limits begin: $K^* 50-60$

16-h day: 27 vehicles for the mean of 339, 117 for the peak of 678 (whole-route reservation)

How many vehicles

Okpo's ordinary days stay below the network-bound regime; its peak falls inside it, where extra vehicles help far less. Published field fleets fall between the model's two lines.

The two yards fall on either side

Yantai moves 23–46 blocks a day with 1–2 vehicles, free of network limits; Okpo's peak is network-bound. The network-bound hypothesis yields a result either way.

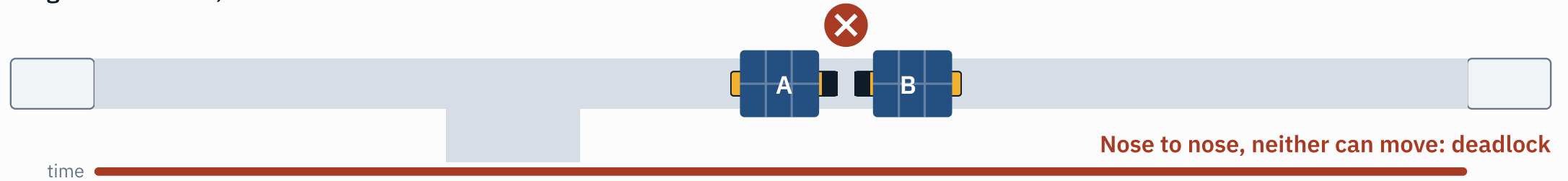
Where it binds: checked on imagery, now a fixed step

The deciding spots (Okpo's dock-1 road, two doors of a Yantai paint shop) sat on cautious modelling choices and changed after an imagery check. Every yard now has its top three shadow-price spots checked.

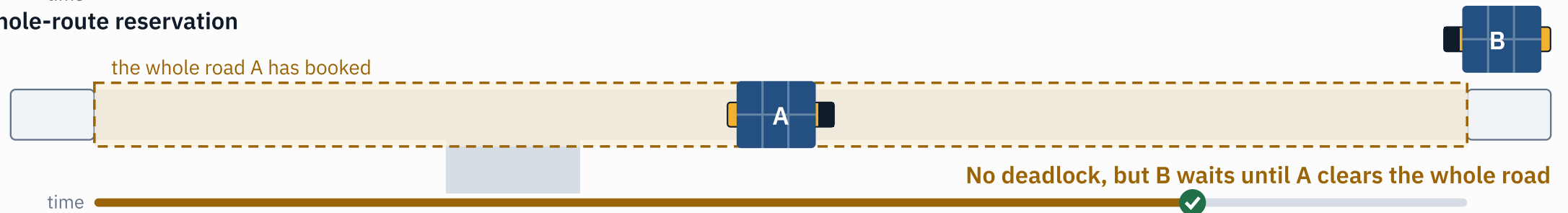
Data: simulation in this study, whole-yard volumes, Okpo main scenario, dock road as one resource; vehicles needed = smallest fleet whose mean throughput reaches 99% of the daily volume; open triangles: not reached by $K = 150$. Field fleets: Park and Seo 2006; Shen et al. 2018. The main basis (16 h or 24 h, mean or peak) follows the rule comparison.

On a single lane: how deadlock arises, and how rules avoid it

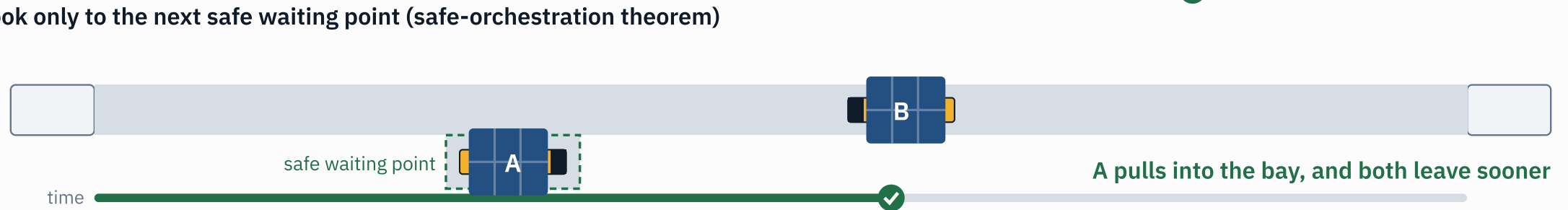
① One segment at a time, no check



② Whole-route reservation

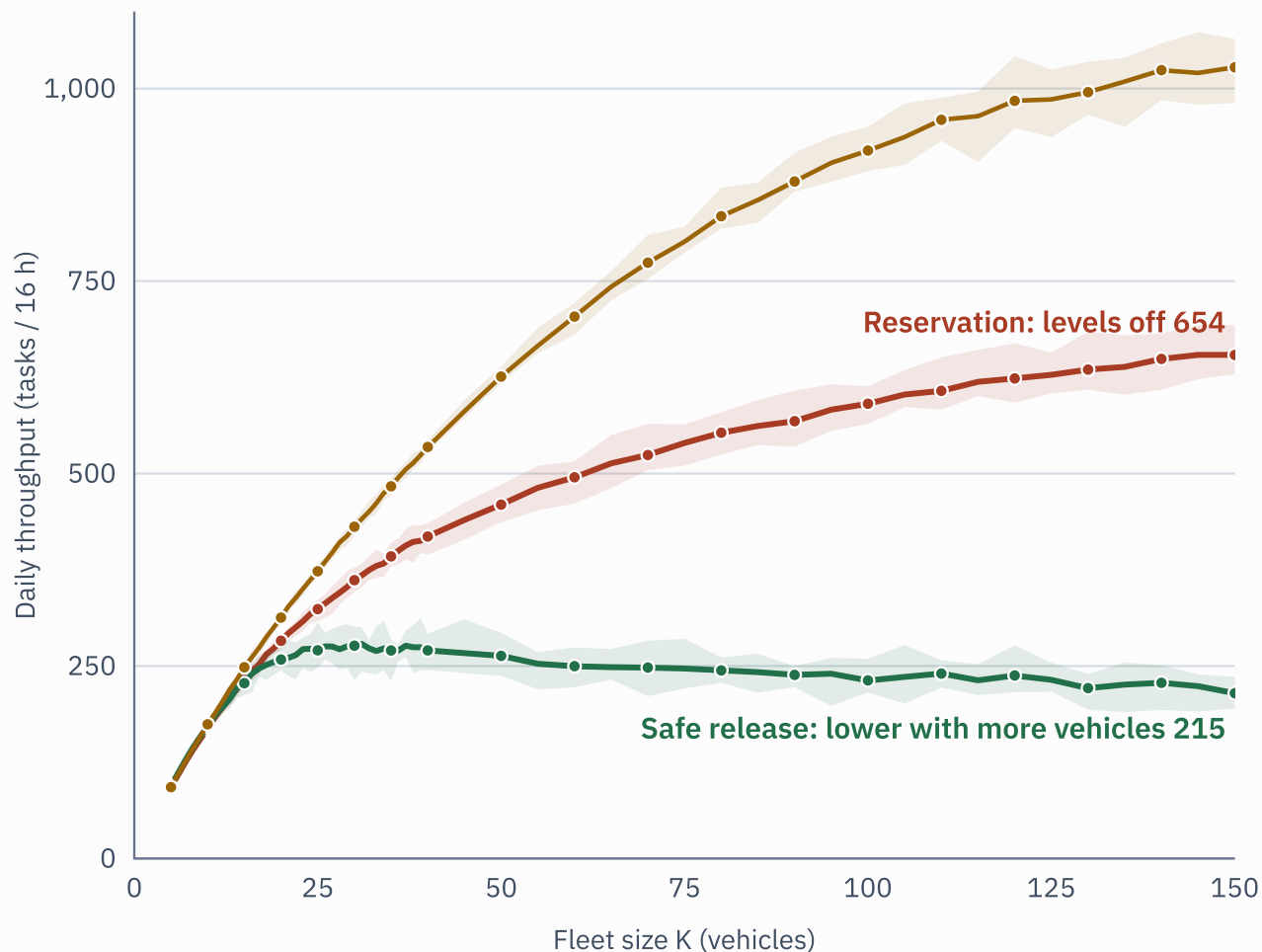


③ Book only to the next safe waiting point (safe-orchestration theorem)



Schematic, not to scale; same speed in all rows, and the bar under each row is the time until both vehicles arrive. ② is the rule behind the current lower edge; ③ is the policy of the safe-orchestration theorem, now tested in the deadlock-free rule comparison; step 3's safe release, which releases only when every vehicle can still finish, sits in between.

Under way: a smarter scheduling rule that still never jams



■ Whole-route reservation ■ Safe release — Segment request (reference)

Why

Step 3's safe release never deadlocks but is over-cautious: from 20 vehicles on it falls below reservation and drops as the fleet grows; about two thirds of its waiting are refusals.

What

Two more tests, releasing if either passes: look a few steps ahead, or book only to the next safe waiting point (③ on the previous slide). The runs first switch to Okpo's new map with its 3 floating docks.

Why it goes first

It sets the bottom of the interval, and with it the line between improving scheduling and changing roads, so it comes before the retrofit simulations.

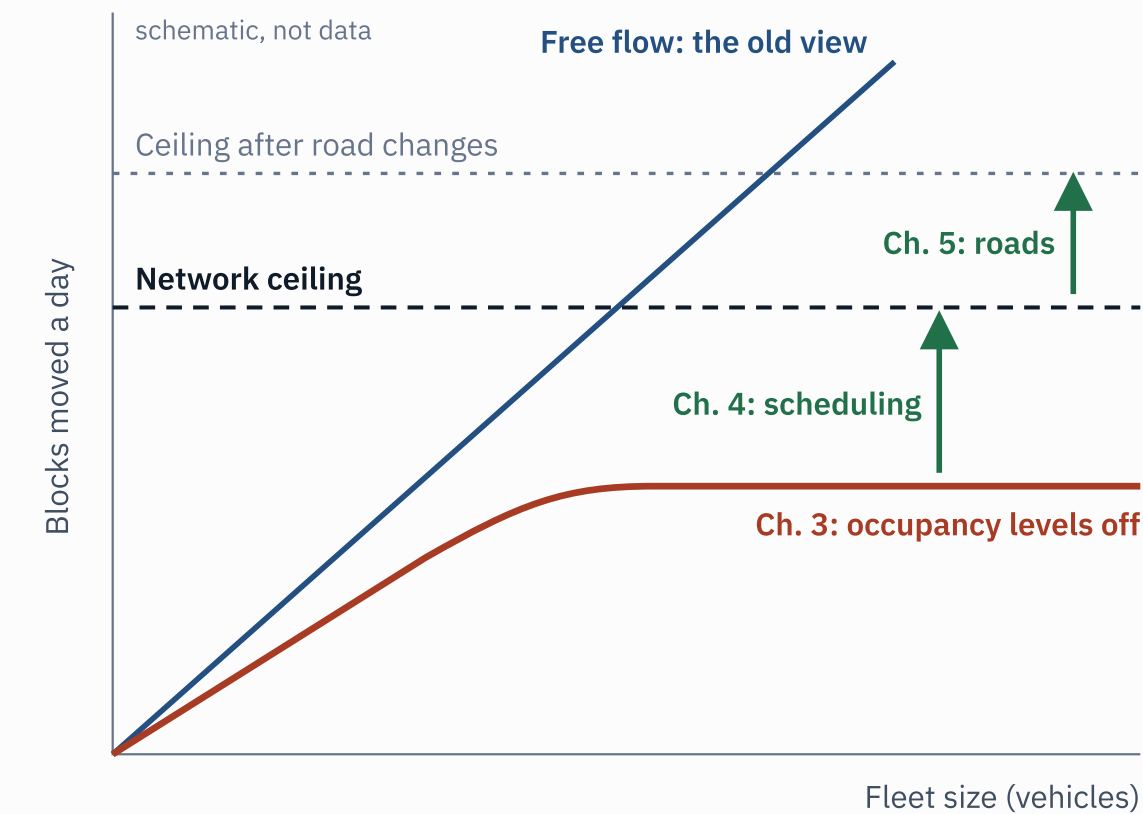
running Also chapter 4's first rule comparison.

Next: vehicles or roads, more yards, the theorems and the first paper

WORK	WHAT IT ANSWERS	WHERE IT GOES
Retrofit simulations	Road splitting, holding bays at the dock, passing bays, widening: how many vehicles each is worth, against the shadow-price prediction	Chapter 5: the first exchange-rate table
Floating docks, finite holding	Giant blocks carried by several vehicles together; with limited holding space, from what fleet size extra vehicles start to hurt	Chapter 5 (vehicle-type hypothesis); chapter 3 (decline hypothesis)
More yards	20 yards on the list already have maps and base images; compute the first group's bounds and test whether the map alone predicts the results	Chapter 6: testing across yards
Theorem proofs	Rewrite the proofs of the network and route-free bounds; prove that free-flow error can be arbitrarily large and that the exchange rate equals the shadow price; word the network-bound hypothesis after the rule comparison	The theoretical core of chapter 3
The first paper	Model and bounds, the capacity interval, the dock mouth, the first exchange-rate table; submitted within 3–4 months	First target: Transportmetrica A or Ocean Engineering

Sources: this study's research plan and candidate yard list; this study's claims and hypotheses.

One figure ties the thesis together



ON THE FIGURE

IN THE THESIS

The straight line

Free flow, the old view (chapter 1)

The curve that levels off

Occupancy: where capacity comes from (chapters 2–3)

Pushing the curve up

Better scheduling (chapter 4)

Raising the ceiling

Roads or vehicles (chapter 5)

One figure per yard

Predicting from the map (chapter 6)

[Open the thesis overview →](#)

[Flag experiment and capacity interval →](#)

In a shipyard, a loaded transporter does not drive on the road; it owns the road. Capacity is capped by the network, not the fleet.

Sources: this study's chapter plan and signature figure. Schematic, not data; the measured curves are in the thesis overview's signature figure.